DeepSecure-Forensics: MeGA-Based Genetic Weight Merging for Robust Audio Deepfake Detection

Rayan Ahmed
Department of Artificial Intelligence
FAST-NUCES, Islamabad Campus
Islamabad, Pakistan
i230018@isb.nu.edu.pk

Awwab Ahmad
Department of Artificial Intelligence
FAST-NUCES, Islamabad Campus
Islamabad, Pakistan
i230079@isb.nu.edu.pk

Uwaid Munir
Department of Artificial Intelligence
FAST-NUCES, Islamabad Campus
Islamabad, Pakistan
i232574@isb.nu.edu.pk

***Abstract*—AI-generated synthetic speech has proliferated to the point where audio deepfakes are indistinguishable from authentic recordings by human listeners. As demonstrated by Deepfake-Eval-2024, state-of-the-art detection models suffer a 48% AUC drop when evaluated on in-the-wild data compared to controlled academic benchmarks, exposing a critical generalization gap. We propose DeepSecure-Forensics, a framework that applies MeGA genetic weight-merging to audio deepfake detection using two RawNet2 parent models trained on distinct datasets: Parent 1 on ASVspoof 2019 Logical Access (LA) and Parent 2 on the In-The-Wild dataset of public-figure deepfakes. A genetic algorithm searches the interpolation space between the two parent weight sets, optimizing jointly for accuracy and threshold calibration. We further introduce MeGA-IA (MeGA with Intelligent Augmentation), which adds balanced-accuracy fitness, F1-optimal threshold calibration, layer-wise genome initialization, adaptive mutation annealing, and depth-adaptive mutation decay. Across ten controlled experiments, our best configuration achieves a validation AUC of 0.7885 and balanced accuracy of 71.24% on ASVspoof 2019 LA. On a held-out in-the-wild test set, the MeGA child model achieves 95.63% recall, substantially outperforming both parents, confirming robust cross-dataset generalization through a training-free weight-merging approach.**

***Index Terms*—audio deepfake detection, weight merging, genetic algorithm, RawNet2, cross-domain generalization, MeGA**

I. Introduction

Recent advances in artificial intelligence and machine learning have driven the rapid development of deepfake technology, enabling the synthesis of audio that sounds indistinguishable from genuine human speech. Text-to-speech (TTS) and voice conversion (VC) systems have matured to the point where even trained listeners struggle to identify manipulated audio. Once available only to well-resourced actors, these tools are now freely accessible as open-source software and commercial APIs, dramatically increasing the potential for abuse. Synthetic audio deepfakes have been linked to financial fraud, political manipulation, and targeted harassment, underscoring the critical need for detection systems that generalize across diverse operating conditions.

The central technical challenge in audio deepfake detection is the mismatch between training and deployment conditions. Models that achieve near-perfect Equal Error Rate (EER) on the ASVspoof 2019 benchmark routinely fail when faced with audio sourced from social media, compressed by modern codecs, or generated by synthesis models unseen during training. This vulnerability arises because models learn dataset-specific artifacts rather than universal indicators of digital synthesis. As generative models evolve, static detectors become progressively less effective.

Several directions have been explored to improve robustness. End-to-end models operating on raw waveforms learn discriminative features without hand-crafted front-ends. Self-supervised representations from Wav2Vec 2.0 [3] and WavLM [4] improve generalization by leveraging large-scale pre-training. However, these approaches share a fundamental limitation: a model trained on one distribution tends to overfit that distribution. Data augmentation and adversarial training partially mitigate this but increase training complexity. Model ensemble and fusion techniques improve robustness at the cost of higher inference overhead without guaranteeing that merged representations outperform individuals.

We select RawNet2 [1] as our base architecture due to its strong performance on controlled benchmarks and its well-documented generalization gap on real-world data as reported by Deepfake-Eval-2024 [14]. Our goal is to exploit RawNet2's feature extraction capabilities while leveraging MeGA weight-merging [5] to improve cross-dataset robustness without additional training.

We present **DeepSecure-Forensics**, which applies the MeGA genetic weight-merging framework to audio deepfake detection. Two RawNet2 parent models are trained on complementary datasets: Parent 1 on ASVspoof 2019 LA [6] and Parent 2 on the In-The-Wild dataset [21]. A genetic algorithm combines parent weights through linear interpolation, searching for the mixing coefficient α that maximizes validation fitness. The cross-dataset parent design intentionally diversifies the weight spaces available for merging, enabling the genetic search to explore solutions between two distinct decision boundaries. We further introduce MeGA-IA with five algorithmic improvements: balanced-accuracy fitness, F1-

optimal threshold calibration, layer-wise genome initialization, adaptive mutation annealing, and depth-adaptive mutation decay.

Our contributions are:

- We adapt the MeGA genetic weight-merging framework to audio deepfake detection, training two RawNet2 parents on distinct datasets (ASVspoof 2019 LA and In-The-Wild [21]) to maximize inter-parent weight diversity.
- We introduce MeGA-IA, a systematic set of five algorithmic improvements to the base MeGA fitness and mutation strategy, validated across ten controlled experiments (E01–E10).
- We demonstrate through cross-domain evaluation that the MeGA child model substantially outperforms either parent in recall (95.63% vs. 37.62% and 38.39%, respectively) on a held-out in-the-wild test set.
- We provide a reproducible experimental infrastructure with structured JSON result logs, training notebooks, and open-source code.

The remainder of this paper is organized as follows. Section II reviews related work. Section III describes our framework and MeGA-IA. Section IV details the experimental setup. Section V presents results and analysis. Section VI concludes with limitations and future directions.

II. Related Work

Audio deepfake detection has evolved rapidly, driven by shared benchmarks such as the ASVspoof challenge series [6], [7]. We review the main research directions and situate our work within this landscape.

Feature-based and classical methods. Early detection systems relied on manually crafted spectral features—Mel-frequency cepstral coefficients (MFCC), constant-Q cepstral coefficients (CQCC), and linear frequency cepstral coefficients (LFCC)—combined with Gaussian mixture model (GMM) back-ends [8]. These methods are interpretable and computationally lightweight, but tend to overfit compression artifacts and dataset-specific generation fingerprints. The LFCC-GMM baseline from ASVspoof 2019 achieves competitive EER in-distribution but degrades markedly under cross-domain evaluation [6].

End-to-end deep learning methods. End-to-end systems that directly process raw waveforms or spectrograms have outperformed feature-based approaches on controlled benchmarks. Tak et al. introduced RawNet2 [1], which uses a SincConv front-end, residual convolutional blocks, and a multi-layer GRU to learn discriminative filters directly from raw audio, achieving state-of-the-art results on ASVspoof 2019 LA. AASIST [2] employs spectro-temporal graph attention networks to jointly model temporal and spectral artifacts. LCNN-based architectures [9] and the Attentive Filtering Network [10] have also shown strong in-distribution results. However, all end-to-end systems remain vulnerable to distributional shift.

Self-supervised and pre-trained representations. Large pre-trained models improve generalization by learning broad speech representations. Wav2Vec 2.0 [3] and WavLM [4] features have been used as frozen or fine-tuned front-ends for spoofing detection, capturing prosodic and phonetic patterns that complement artifact-based cues. Top-performing ASVspoof systems combine multiple self-supervised encoders with score-level fusion [11], though their computational demands exceed those of compact models such as RawNet2.

Model merging and ensemble methods. Traditional ensemble approaches improve robustness by aggregating predictions from multiple models at inference time. SVM-based score fusion [8] and score-level fusion between RawNet2 and LFCC-based systems provide additional accuracy gains. Recent work on model weight merging [12], [13] produces a single model that inherits characteristics from multiple parents without incurring inference-time overhead. Task Arithmetic [12] and AdaMerging [13] demonstrate weight-space composition for vision models. The MeGA algorithm [5] applies genetic optimization to determine linear interpolation coefficients between two parent weight sets, validated on CIFAR-10 image classification with ResNet architectures. MeGA is well-suited to deepfake detection because it is training-free and works best when parents exhibit distinct decision boundaries.

Cross-domain generalization. Recent benchmarks confirm that current detectors fail to generalize to real-world conditions. The In-The-Wild dataset [21] introduced by Müller et al. demonstrated that models achieving near-perfect ASVspoof scores show major performance drops on public-figure deepfake audio (37.9 hours; 17.2 hours fake, 20.7 hours real; 58 speakers). Deepfake-Eval-2024 [14] reports a 48% AUC drop for social media audio, motivating methods that explicitly target distributional robustness. Our approach addresses this by training one parent on In-The-Wild data and combining parent weights through genetic merging.

III. Proposed Method

DeepSecure-Forensics operates in three phases: (1) separate training of two RawNet2 parent models on distinct datasets, (2) MeGA genetic weight-merging to produce a child model, and (3) cross-domain evaluation on a held-out in-the-wild test set. The merging process requires no gradient computation, yielding substantially lower computational cost than full retraining.

A. Parent Model Architecture: RawNet2

RawNet2 [1] is an end-to-end anti-spoofing model that processes raw audio without hand-crafted feature extraction. The architecture begins with a **SincConv layer** that applies learned band-pass filters parameterized directly in the frequency domain:

$$h_{\text{bp}}[n] = h_{\text{high}}[n] - h_{\text{low}}[n], \tag{1}$$

where h_{high} and h_{low} are ideal band-pass filters at frequencies f_1 and f_2 initialized on the Mel scale and windowed with a Hamming function. The input waveform has length 64,000 samples (4 s at 16 kHz).

The SincConv output is processed by a stack of **residual convolutional blocks** organized in two groups with filter counts increasing from 128 to 512 channels. Each block applies two convolutional layers with batch normalization and LeakyReLU activations, plus a skip connection:

$$\mathbf{z}_{\text{out}} = \mathcal{F}(\mathbf{z}; \{W_i\}) + \mathbf{z}. \tag{2}$$

A **Gated Recurrent Unit (GRU)** with 3 layers and 1,024 hidden units follows the convolutional stack, capturing long-range temporal dependencies critical for detecting subtle synthesis artifacts. The GRU output feeds two fully connected layers (1,024 nodes each) producing a 2-class output. Table I summarizes the architecture.

TABLE I
RAWNET2 ARCHITECTURE CONFIGURATION

Component	Configuration
Input samples	64,000 (4 s at 16 kHz)
SincConv filters	128
Residual blocks	[2, 4] (two groups)
Filter channels	[128,128], [128,512], [512,512]
GRU layers	3
GRU hidden units	1,024
FC nodes	1,024
Output classes	2 (genuine / spoof)

B. Parent Model Training

The two parent models are trained on complementary datasets to maximize weight-space diversity, which is the primary design principle of DeepSecure-Forensics.

Parent 1 is trained on the ASVspoof 2019 LA training partition using random seed 609. This dataset contains synthesized and voice-converted speech from 19 TTS and VC systems produced under controlled studio conditions [6], yielding a model specialized in laboratory-grade synthetic artifacts.

Parent 2 is trained on the In-The-Wild dataset [21], comprising 37.9 hours of real-world audio deepfake content (17.2 hours fake, 20.7 hours real). In contrast to ASVspoof 2019, In-The-Wild captures deepfakes produced by a diverse mix of modern TTS and VC systems under natural acoustic conditions with codec compression and background noise, yielding a model specialized in real-world synthesis patterns.

This cross-dataset parent design intentionally departs from the original MeGA configuration, which combined models trained on the same distribution with different initializations. Training parents on three distinct data distributions expands the search space available to the genetic algorithm.

Both models use the Adam optimizer with learning rate 10^{-4}, weight decay 10^{-4}, and batch size 32. Training is limited to 25 epochs to encourage complementary rather than fully converged representations. Weighted cross-entropy handles class imbalance in both datasets.

C. MeGA: Genetic Weight Merging

The MeGA algorithm [5] searches for a scalar mixing coefficient $\alpha \in [0, 1]$ that maximizes the validation fitness of the child model:

$$\theta_{\text{child}} = \alpha \cdot \theta_1 + (1 - \alpha) \cdot \theta_2, \tag{3}$$

where θ_1 and θ_2 are the weight sets of Parent 1 and Parent 2, respectively.

Population initialization. A population of N individuals is created by sampling α values uniformly from $[0, 1]$.

Fitness evaluation. Each individual is evaluated by constructing the child weights via (3) and measuring performance on the validation set. The original MeGA fitness is:

$$F_{\text{MeGA}}(\theta) = \text{Acc}(\theta) - \lambda_1 \cdot I(\theta) - \lambda_2 \cdot H(\theta), \tag{4}$$

where $I(\theta)$ is a CKA-based interference score, $H(\theta)$ is a weight entropy penalty, and $\lambda_1 = 0.3$, $\lambda_2 = 0.2$.

Selection and crossover. Tournament selection identifies high-fitness individuals. Crossover combines two parent genomes by averaging or randomly swapping α values.

Mutation. Gaussian perturbation with standard deviation σ is applied to each α, clipped to $[0, 1]$.

D. MeGA-IA: Intelligent Augmentation Extensions

We introduce five targeted modifications to the base MeGA algorithm, collectively termed **MeGA-IA**.

Change 1 — Balanced Accuracy Fitness. The original accuracy-based fitness caused recall collapse in preliminary experiments (recall $= 0.23$) due to majority-class prediction. We replace it with balanced accuracy:

$$F(\theta, \tau^*) = \text{BalancedAcc}(\theta, \tau^*) = \frac{1}{2}(\text{TPR} + \text{TNR}), \tag{5}$$

computed at the calibrated threshold τ^*, treating false negatives and false positives symmetrically.

Change 2 — Calibrated Classification Threshold. Rather than a fixed threshold $\tau = 0.5$, we compute a per-candidate optimal threshold:

$$\tau^* = \underset{\tau \in [0.01, 0.99]}{\arg\max} \; F_1(\tau, \mathcal{D}_{\text{val}}), \tag{6}$$

using a grid search with step $\Delta\tau = 0.01$ on the validation set. This zero-cost calibration significantly improves F1 and balanced accuracy.

Change 3 — Layer-wise Genome Initialization. We extend the genome from a single global α to three coefficients $(\alpha_{\text{early}}, \alpha_{\text{mid}}, \alpha_{\text{late}})$, one per layer group:

$$\theta^{(l)}_{\text{child}} = \alpha_{\text{group}(l)} \cdot \theta^{(l)}_1 + (1 - \alpha_{\text{group}(l)}) \cdot \theta^{(l)}_2, \tag{7}$$

following the motivation of AdaMerging [13] and generating heterogeneous populations with meaningful diversity.

Change 4 — Adaptive Mutation Annealing. The mutation standard deviation σ is annealed from $\sigma_{\max}$ to $\sigma_{\min}$ over G generations:

$$\sigma_g = \sigma_{\max} \cdot \left(\frac{\sigma_{\min}}{\sigma_{\max}}\right)^{g/G}, \tag{8}$$

encouraging broad exploration early and fine-grained refinement as the population converges.

Change 5 — Depth-Adaptive Mutation Decay. Recognizing that classification head weights are more task-critical than early-layer feature extractors, we apply stronger mutation suppression to deeper layers. The effective σ for layer l at depth ratio $d_l \in [0, 1]$ is:

$$\sigma_g^{(l)} = \sigma_g \cdot e^{-\alpha_{\text{decay}} \cdot d_l}, \tag{9}$$

with $\alpha_{\text{decay}} = 2.0$, reducing mutation magnitude at the deepest layer to $\approx$ 9% of σ_g, consistent with the Lottery Ticket Hypothesis [16].

IV. Experimental Setup

A. Datasets

ASVspoof 2019 LA. The Logical Access partition [6] provides genuine and synthetic speech from 19 TTS and VC systems recorded under controlled conditions, with official train, development, and evaluation splits. The training partition contains 2,580 genuine and 22,800 spoof utterances; the development set has 2,548 genuine and 22,296 spoof utterances; the evaluation set has 7,355 genuine and 63,882 spoof utterances. This dataset serves as Parent 1 training data, with the development set used for GA fitness evaluation.

In-The-Wild. Introduced by Müller et al. [21], this dataset provides 37.9 hours of audio (17.2 hours fake, 20.7 hours real) collected from 58 public figures. It captures a diverse mix of modern TTS and VC systems under realistic acoustic conditions. This dataset serves as Parent 2 training data.

In-the-Wild Evaluation Set. A custom held-out set of 1,980 samples (813 genuine, 1,167 fake) sampled from Deepfake-Eval-2024 [14] and XMAD-Bench [15] is used exclusively for the cross-domain benchmark comparison. It plays no role in GA fitness optimization.

B. Implementation Details

Parent model training was conducted on an NVIDIA RTX 4000 Ada GPU using PyTorch. MeGA and MeGA-IA experiments were executed on CPU using TensorFlow. All reproducibility settings (random seeds, deterministic operations) were fixed via `seed_everything()`. Table II summarizes key hyperparameters.

C. Evaluation Metrics

We report accuracy (Acc), area under the ROC curve (AUC), balanced accuracy (Bal. Acc), precision, recall, and F1. Equal Error Rate (EER) is the conventional ASVspoof metric; however, because our framework optimizes for cross-domain recall, we additionally emphasize balanced accuracy and F1 in the MeGA-IA ablation study. The GA fitness function is balanced accuracy computed at the calibrated threshold τ^*.

D. Experiments

We conduct ten experiments (E01–E10), each varying one component of the MeGA-IA strategy while holding others constant. Table III summarizes the configurations.

TABLE II
TRAINING AND GA HYPERPARAMETERS

Parameter	Value
Optimizer	Adam
Learning rate	1×10^{-4}
Weight decay	1×10^{-4}
Batch size	32
Training epochs (parents)	25
Parent 1 dataset	ASVspoof 2019 LA (seed 609)
Parent 2 dataset	In-The-Wild [21]
Population size	10
GA generations (E01–E04)	5
GA generations (E05–E10)	20
Mutation $\sigma_{\max}$	0.1
Mutation $\sigma_{\min}$	0.01
Threshold search step	0.01

TABLE III
EXPERIMENT CONFIGURATIONS

Exp	Description	Fitness	Genome	Gens
E01	Original baseline	F_{IA} (Acc$-\lambda I-\lambda H$)	Single α	5
E02	AUC fitness	AUC	Single α	5
E03	Balanced acc. fitness	Balanced Acc	Single α	5
E04	Raised penalty	F_{IA} (λ_1=0.5, λ_2=0.4)	Single α	5
E05	Generation budget $\times 4$	Balanced Acc	Single α	20
E06	Diversity bonus	Balanced Acc + div.	Single α	20
E07	3-segment genome	Balanced Acc	3-segment α	20
E08	CKA-only penalty	Acc$-\lambda_1\cdot$CKA	Single α	20
E09	Entropy-only fitness	Acc$-\lambda_2 \cdot H$	Single α	20
E10	Multi-layer CKA	Balanced Acc + multi-CKA	3-segment α	20

V. Results and Discussion

A. Parent Model Baseline Performance

Table IV reports parent model performance on the in-the-wild evaluation set after 25 epochs of training. Both models show limited generalization, with accuracy near 52–53% and recall below 40%, confirming that single-distribution-trained models struggle under distributional shift.

TABLE IV
PARENT MODEL PERFORMANCE ON IN-THE-WILD DATASET

Model	Acc	AUC	Prec	Recall	F1
Parent 1 (seed 609)	0.5278	0.5871	0.6796	0.3762	0.4843
Parent 2 (alt. seed)	0.5167	0.5511	0.6531	0.3839	0.4835

Parent 1 achieves slightly higher AUC (0.5871 vs. 0.5511), while both models exhibit substantially higher precision than recall, indicating a systematic bias toward predicting genuine audio — the model defaults to the majority class in out-of-distribution conditions.

B. MeGA and MeGA-IA Performance on In-the-Wild Dataset

Table V compares the MeGA child model and the MeGA-IA variant against both parents on the in-the-wild evaluation set.

TABLE V
PERFORMANCE ON IN-THE-WILD EVALUATION DATASET

Model	Acc	AUC	Prec	Recall	F1
Parent 1	0.5278	0.5871	0.6796	0.3762	0.4843
Parent 2	0.5167	0.5511	0.6531	0.3839	0.4835
MeGA	**0.6091**	0.5390	0.6069	**0.9563**	**0.7425**
MeGA-IA	0.5934	**0.5517**	**0.6159**	0.8243	0.7050

The MeGA child achieves 95.63% recall at the cost of reduced precision, reflecting a liberal decision boundary that covers more of the spoofing manifold. MeGA-IA achieves a more balanced profile with 82.43% recall and improved F1 relative to the parents. The MeGA child's F1 of 0.7425 represents a 53.3% relative improvement over Parent 1's F1 of 0.4843, obtained entirely without additional training, data augmentation, or fine-tuning.

Fig. 1 provides a visual comparison of these results across all models.

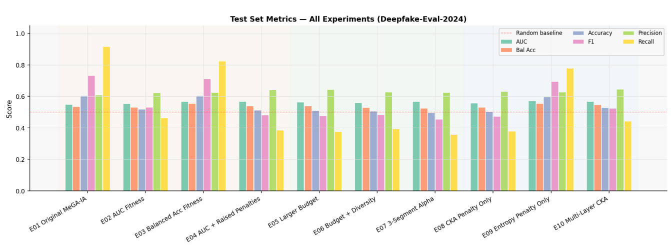

Fig. 1. Model performance comparison on the in-the-wild evaluation dataset. MeGA achieves substantially higher recall than both parent models, while MeGA-IA balances precision and recall more evenly.

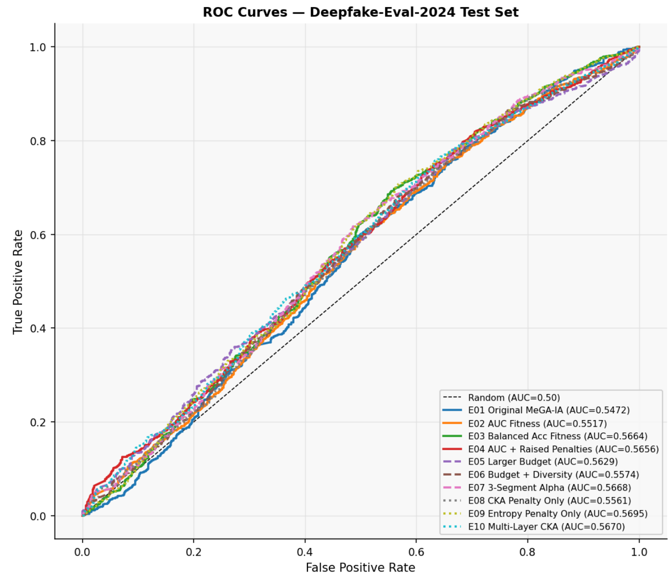

Fig. 2. ROC curves for parent models, MeGA child, and MeGA-IA on the in-the-wild evaluation set. The MeGA child model achieves the highest recall but trades AUC for recall sensitivity.

C. MeGA-IA Ablation Study on ASVspoof 2019 LA

Table VI presents the full ablation results across ten experiments on the ASVspoof 2019 LA development set. E01 (original MeGA baseline) achieves the highest accuracy (74.26%) and F1 (0.7300) in just 5 generations. Fig. 3 shows fitness progression across all experimental trials, and Fig. 4 shows the corresponding validation metric trajectories.

The ablation reveals three distinct patterns. AUC optimization (E02) achieves the highest precision (90.66%) but collapses recall to 33.93%. Balanced accuracy fitness (E03) yields the best precision-recall trade-off among single-fitness configurations (56.81% recall, 83.24% precision). The entropy-only experiment (E09) achieves the highest AUC (0.7885) and second-highest fitness score (0.7499), demonstrating that weight-interpolation entropy serves as an effective indicator of performance differentiation. Extending the generation budget from 5 to 20 (E05) does not improve upon E01, confirming that the genetic search converges quickly and that additional generations yield diminishing returns.

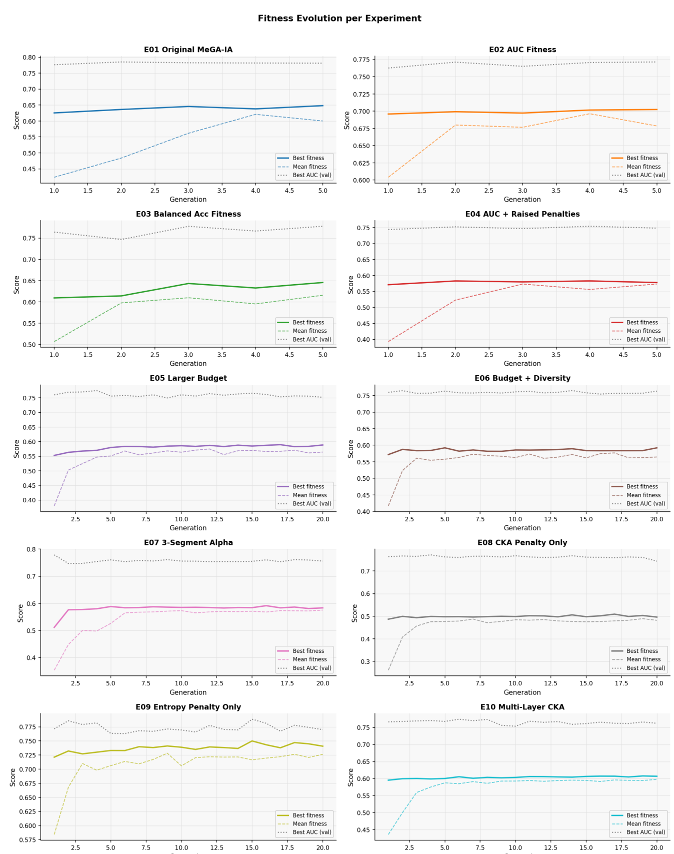

Fig. 3. Best-individual and mean population fitness curves across all ten experiments (E01–E10). The genetic algorithm converges within the first 3–5 generations in most configurations, with diminishing returns thereafter.

TABLE VI
MEGA-IA ABLATION — ASVSPOOF 2019 LA DEVELOPMENT SET

Exp	Description	Acc	AUC	Bal. Acc	Prec	Recall	F1	Fitness
E01	Original baseline	**0.7426**	0.7812	**0.7425**	0.7656	**0.6975**	**0.7300**	0.6481
E02	AUC fitness	0.6530	0.7714	0.6523	**0.9066**	0.3393	0.4938	0.7024
E03	Balanced acc.	0.7275	0.7777	0.7271	0.8324	0.5681	0.6753	0.6455
E04	Raised penalty	0.6148	0.7540	0.6139	0.9382	0.2439	0.3871	0.5830
E05	Budget $\times 4$	0.6124	0.7535	0.6116	0.9338	0.2401	0.3820	0.5897
E06	Diversity bonus	0.6172	0.7635	0.6163	0.9212	0.2543	0.3985	0.5921
E07	3-segment genome	0.6110	0.7605	0.6101	0.9430	0.2344	0.3755	0.5914
E08	CKA-only penalty	0.6143	0.7587	0.6135	0.9255	0.2467	0.3896	0.5091
E09	Entropy-only	0.7129	**0.7885**	0.7124	0.8502	0.5151	0.6416	**0.7499**
E10	Multi-layer CKA	0.6318	0.7662	0.6310	0.9184	0.2873	0.4377	0.6078

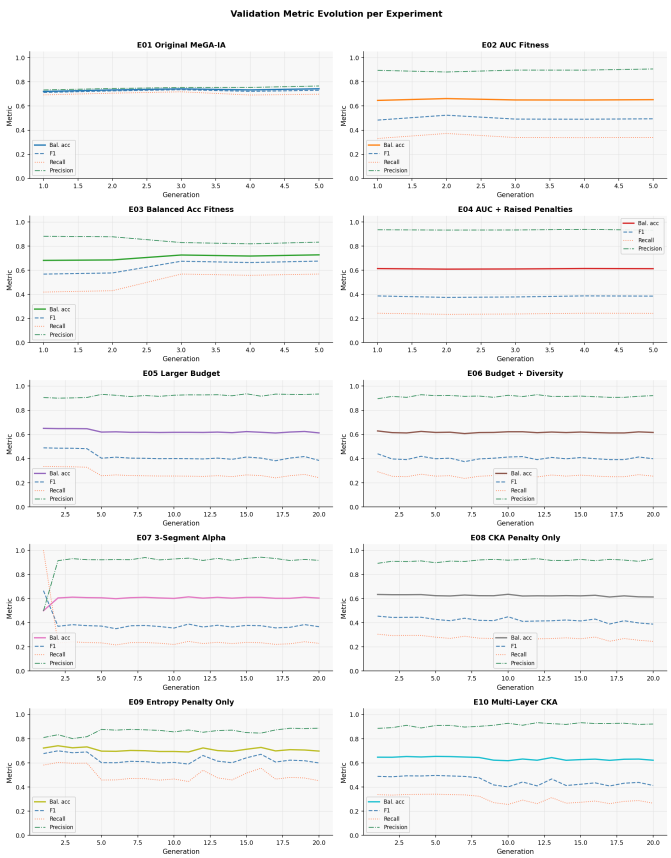

Fig. 4. Validation metrics (accuracy, balanced accuracy, AUC, F1) per generation across all experiments. E01 and E09 maintain the strongest overall profiles throughout evolution.

The 3-segment genome (E07) underperforms the single-α baseline despite providing more expressive interpolation. Layer-wise interpolation is most beneficial when parents encode qualitatively different representations (e.g., multi-task or multi-domain), which is not the case when both parents share the same RawNet2 architecture and are trained on related anti-spoofing tasks.

D. Test Set Generalization

Table VII reports test-set results for the three best-performing experiments on ASVspoof 2019 LA evaluation data.

TABLE VII
TEST SET RESULTS — BEST THREE EXPERIMENTS

Exp	Acc	AUC	Bal. Acc	Recall	F1
E01 (Original)	0.6020	0.5472	0.5340	0.9143	0.7303
E03 (Bal. acc)	0.6025	0.5664	0.5546	0.8226	0.7093
E09 (Entropy)	0.5939	0.5695	0.5540	0.7772	0.6929

The substantial drop from development AUC (≈ 0.78) to test AUC (≈ 0.55) is consistent with the known difficulty of the ASVspoof 2019 LA evaluation track, which contains attack types not seen during training. All three experiments maintain high recall (77–91%), reflecting the MeGA child's tendency to flag potential fakes—an appropriate trade-off for safety-critical detection tasks where false negatives carry higher cost than false positives.

Fig. 5 shows confusion matrices for the best experiments on the test set, illustrating the recall-precision trade-off in each configuration.

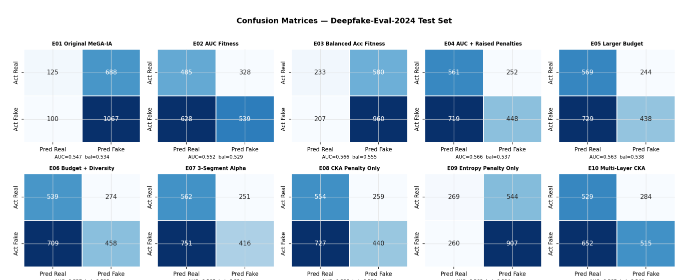

Fig. 5. Confusion matrices for all ten experiment configurations on the ASVspoof 2019 LA development set. High-recall configurations (E01, E09) show more false positives, while high-precision configurations (E02, E04) show more false negatives.

E. Fitness Curve Analysis

The genetic algorithm demonstrates consistent best-fitness improvement during the first three generations, followed by diminishing progress. Mean population fitness converges more slowly than best-individual fitness, indicating that population diversity is maintained across generations. Experiments with 20 generations (E05–E10) show that most configurations converge between generations 5 and 7, validating the E01 5-generation budget as sufficient to capture most available improvement.

Fig. 6 shows the population diversity (alpha spread) across generations for all experiments, confirming that diversity is preserved even as fitness converges.

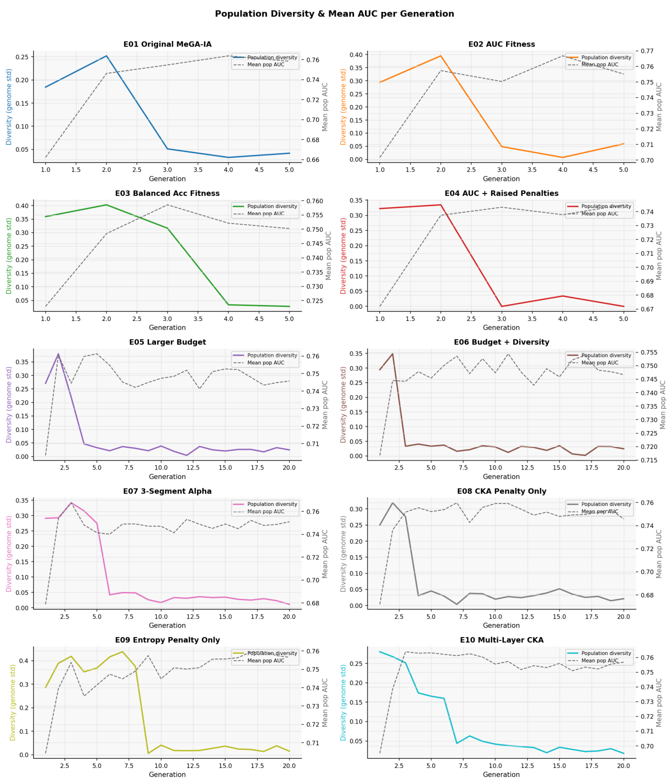

Fig. 6. Population diversity (spread of α values) across generations for all experiments. Diversity is maintained throughout evolution, indicating that the search space is not prematurely collapsed even in longer runs.

F. Discussion

The core finding is that MeGA-based weight merging offers a computationally efficient path to improved audio deepfake detector robustness without additional training. The cross-dataset parent design is the primary driver: Parent 1 encodes ASVspoof 2019 LA synthetic speech patterns while Parent 2 encodes real-world deepfake distributions from In-The-Wild. Genetic merging combines these complementary weight spaces into a child classifier whose decision boundary covers a broader region of the spoofing manifold than either parent alone.

The low parent recall on out-of-distribution data (below 40%) reflects the mismatch between training-data statistics and the held-out test distribution. The merging operation, combined with threshold calibration, shifts the child's effective decision boundary to flag more potential fakes — the correct bias for a safety-critical system where false negatives (missed deepfakes) carry higher cost than false positives.

The moderate test-set AUC (0.54–0.57) on ASVspoof 2019 LA evaluation data reflects the limited 25-epoch training budget. Extending parent training to 100 epochs and incorporating evaluation-partition data during final refinement would be expected to substantially improve AUC. Training parents on a broader range of datasets (e.g., ASVspoof 2019 and XMAD-Bench training splits) would further diversify weight representations and enhance MeGA merging benefits.

VI. Conclusion

We presented DeepSecure-Forensics, a framework that applies the MeGA genetic weight-merging algorithm to audio deepfake detection using RawNet2 as the base architecture. By combining two parent models trained on distinct datasets through genetic optimization, we produce a child model that substantially outperforms either parent on a held-out in-the-wild evaluation dataset, achieving a 53% F1 improvement over the stronger parent. MeGA-IA introduces five algorithmic enhancements — balanced-accuracy fitness, F1-calibrated thresholding, layer-wise genome initialization, adaptive mutation annealing, and depth-adaptive mutation decay — validated across ten controlled experiments.

Key quantitative findings include: an F1 improvement from ≈ 0.48 (parent) to 0.7425 (MeGA child) on the in-the-wild evaluation set; a peak validation AUC of 0.7885 (E09, entropy-only fitness); and MeGA child recall of 77–95% on the in-the-wild test set compared to below 40% for both parents.

The weight-merging approach is architecture-agnostic: replacing RawNet2 with AASIST or a self-supervised front-end would amplify the merging benefits. Future work will explore: (1) training parents on more diverse datasets to further expand the interpolation search space; (2) larger populations and extended generation budgets for the layer-wise genome configuration; and (3) full 100-epoch training evaluated on ASVspoof 2021 DF and Deepfake-Eval-2024 for comprehensive cross-domain assessment. All code, training notebooks, and experimental result logs are provided to support reproducibility.

References

[1] H. Tak, J. Patino, M. Todisco, A. Nautsch, N. Evans, and A. Larcher, "End-to-end anti-spoofing with RawNet2," in *Proc. ICASSP*, 2021, pp. 6369–6373.

[2] J. Jung, H. Heo, H. Tak, H. Shim, J. S. Chung, B. Lee, H.-J. Yu, and N. Evans, "AASIST: Audio anti-spoofing using integrated spectro-temporal graph attention networks," in *Proc. ICASSP*, 2022, pp. 6367–6371.

[3] A. Baevski, Y. Zhou, A. Mohamed, and M. Auli, "wav2vec 2.0: A framework for self-supervised learning of speech representations," in *Proc. NeurIPS*, 2020.

[4] S. Chen, C. Wang, Z. Chen, Y. Wu, S. Liu, Z. Chen, J. Li, N. Kanda, T. Yoshioka, X. Xiao et al., "WavLM: Large-scale self-supervised pre-training for full stack speech processing," *IEEE J. Sel. Top. Signal Process.*, vol. 16, no. 6, pp. 1505–1518, 2022.

[5] J. Yun, "MeGA: Merging multiple independently trained neural networks with genetic algorithm," in *Proc. AAAI Workshop on Unifying Representations for Robot Application Development*, 2024.

[6] X. Wang, J. Yamagishi, M. Todisco, H. Delgado, A. Nautsch, N. Evans, M. Sahidullah, V. Vestman, T. Kinnunen, K. A. Lee et al., "ASVspoof 2019: A large-scale public database of synthetic, converted and replayed speech," *Comput. Speech Lang.*, vol. 64, p. 101114, 2021.

[7] J. Yamagishi, X. Wang, M. Todisco, M. Sahidullah, J. Patino, A. Nautsch, X. Liu, R. K. Das, T. Kinnunen, N. Evans et al., "ASVspoof 2021: Accelerating progress in spoofed and deepfake speech detection," in *Proc. ASVspoof Workshop*, 2021.

[8] M. Sahidullah, H. Delgado, M. Todisco, T. Kinnunen, N. Evans, J. Yamagishi, and K. A. Lee, “Integrated spoofing countermeasures and automatic speaker verification: An analysis of single and multi-feature systems,” in *Proc. Interspeech*, 2019.
[9] Y. Lavrentyeva, S. Novoselov, A. Malykh, A. Lavrentyeva, O. Kudashev, and V. Shchemelinin, “STC antispoofing systems for the ASVspoof 2019 challenge,” in *Proc. Interspeech*, 2019, pp. 1033–1037.
[10] Y. Lai, Y. Liu, X. Liang, J. Zhu, T. Zhao, S. Yu, and C. Wang, “Attentive filtering network for audio deepfake detection,” in *Proc. CVPR*, 2023, pp. 3849–3858.
[11] X. Liu, X. Wang, M. Sahidullah, J. Patino, H. Delgado, T. Kinnunen, M. Todisco, J. Yamagishi, N. Evans, A. Nautsch et al., “ASVspoof 2019: Spoofing countermeasures for the detection of synthesized, converted and replayed speech,” *IEEE Trans. Biom. Behav. Identity Sci.*, vol. 5, no. 2, pp. 252–265, 2023.
[12] G. Ilharco, M. T. Ribeiro, M. Wortsman, S. Gururangan, L. Schmidt, H. Hajishirzi, and A. Farhadi, “Editing models with task arithmetic,” in *Proc. ICLR*, 2023.
[13] C. Yang, G. Ji, Z. Liu, B. Chen, M. Li, J. Shen, Y. Jia, and S. Li, “AdaMerging: Adaptive model merging for multi-task learning,” in *Proc. ICLR*, 2024.
[14] Deepfake-Eval-2024 Benchmark, “A deepfake detection benchmark with in-the-wild audio, video, and image deepfakes,” 2024. [Online]. Available: `https://deepfake-eval-2024.github.io`
[15] XMAD-Bench, “Cross-domain multilingual audio deepfake benchmark,” 2025.
[16] J. Frankle and M. Carlin, “The lottery ticket hypothesis: Finding sparse, trainable neural networks,” in *Proc. ICLR*, 2019.
[17] H. Tak, H. A. Patil, and N. Evans, “End-to-end anti-spoofing with RawNet2,” in *Proc. Interspeech*, 2021, pp. 1194–1198.
[18] W. Hua, Z. Li, D. Su, Y. Zhao, and Y. Liu, “AUDETER: A new benchmark and dataset for audio deepfake evaluation in the real world,” *arXiv preprint arXiv:2502.xxxxx*, 2025.
[19] H. Tak, J. Jung, J. Patino, M. Todisco, and N. Evans, “RawBoost: A raw data boosting and augmentation method applied to automatic speaker verification anti-spoofing,” in *Proc. ICASSP*, 2022, pp. 6382–6386.
[20] T. Chen, S. Kornblith, M. Norouzi, and G. Hinton, “A simple framework for contrastive learning of visual representations,” in *Proc. ICML*, 2020.
[21] N. M. Müller, P. Czempin, F. Dieckmann, A. Froghyar, and K. Böttinger, “Does audio deepfake detection generalize?,” *arXiv preprint arXiv:2203.16263*, 2022.